



OPEN Machine learning prediction of future land surface temperature from SAR optical fusion under urban expansion in Changsha, China

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In the context of growing urbanization and persistent cloud contamination in optical remote sensing, reliable large-scale land surface temperature (LST) monitoring in subtropical regions remains a significant challenge. To address this issue, this study develops an innovative SAR–optical collaborative framework that integrates Sentinel-1 dual-polarization features with Landsat-8 observations for improving land use and land cover (LULC) classification, restoring cloud-covered areas, and enabling high-quality reconstruction of cloud-free LST. Building on this foundation, future LULC dynamics were projected using the PLUS model, and LST variations were predicted with the XGBoost algorithm, which enabled quantification of LULC-specific contributions to urban thermal change. The prediction model achieved high accuracy (RMSE = 0.9940 °C, MAE = 0.4714 °C, $R = 0.9819$), underscoring the robustness of SAR–optical integration for LST reconstruction. The results further revealed a strong synchrony between built-up expansion and the increase in LST. Between 2024 and 2030, built-up land is projected to expand by 10.8%, accompanied by a 0.18% increase in extreme high-temperature areas. Overall, the proposed cloud-resilient LST retrieval and prediction framework offers practical value for urban climate adaptation, providing quantitative evidence to support heat mitigation planning, the optimization of green-blue infrastructure, and resilience-oriented spatial development.

Keywords Land use and land cover, Land surface temperature, Synthetic aperture radar, PLUS model, Machine learning

Rapid urbanization has driven extensive transformations in land use and land cover (LULC), intensifying the urban heat island (UHI) effect through the widespread replacement of natural surfaces with impervious materials^{1–4}. The resulting rise in land surface temperature (LST) poses significant challenges for public health, energy demand, and urban sustainability^{5,6}. Although satellite-based thermal infrared (TIR) observations from sensors such as Landsat play a central role in monitoring urban thermal dynamics, their effectiveness is substantially constrained in humid subtropical regions. Persistent cloud cover frequently obscures optical measurements, generating substantial spatiotemporal discontinuities in LST records and hindering reliable long-term thermal assessments^{7–11}. Changsha—a rapidly expanding subtropical megacity—exemplifies these challenges, providing an ideal setting for developing cloud-resilient LST monitoring approaches.

The consequences of cloud-induced data gaps extend beyond incomplete observations to fundamentally limit the accuracy of predictive LST modeling. Forecasting future thermal conditions requires continuous and high-quality historical time series, yet optical satellite data in subtropical regions rarely meet this requirement^{12,13}. Although machine-learning approaches such as Random Forest and XGBoost have achieved notable success in capturing nonlinear relationships between LST and its driving factors, their performance deteriorates when trained on fragmented or heterogeneous datasets^{14–18}. As a result, existing models often struggle to generate unbiased and temporally stable predictions in cloud-prone, rapidly urbanizing cities such as Changsha^{19–22}.

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These limitations underscore the need for alternative data sources that can provide consistent, temperature-relevant information independent of atmospheric conditions.

Synthetic aperture radar (SAR), with its all-weather and day–night imaging capabilities, offers a promising alternative for overcoming the limitations of optical and thermal observations. Although SAR backscattering—particularly from dual-polarization or polarimetric modes—has been widely applied in LULC classification and urban morphology assessment (e.g., through Freeman or Yamaguchi decompositions), its direct use for LST reconstruction and prediction remains largely underexplored^{23–25}. Existing studies predominantly rely on SAR-derived structural or textural indicators, which are only indirectly related to surface thermal properties. Parallel advances in optical-based reconstruction methods, including spatiotemporal fusion, temporal interpolation, and machine-learning regression, have improved the continuity of LST datasets^{26,27}. Yet, they remain fundamentally constrained in persistently cloudy regions. Spatial interpolation degrades sharply in heterogeneous urban environments, where infrequent cloud-free observations limit the effectiveness of temporal methods and fusion strategies, which can propagate cloud-related errors from the source data^{28,29}. Even deep-learning approaches are hindered by the scarcity of high-quality training samples in cloud-obscured areas^{30,31}. These limitations collectively underscore the need to explore SAR as a complementary data source for generating reliable and cloud-resilient thermal information.

Recent studies from subtropical and rapidly urbanizing cities further illustrate the complex and context-dependent drivers of urban thermal dynamics. In Bangkok, accelerated urban expansion has intensified LST, while remaining green corridors provide localized cooling effects³². In Guangzhou, variations in building density, height, and green-space configuration have been shown to regulate UHI intensity, highlighting the role of three-dimensional urban morphology³³. Similar patterns are observed in Wuhan, where increased building height is associated with reduced daytime LST, suggesting a vertical cooling effect³⁴. In Ahmedabad, long-term simulations reveal that farmland-to-built-up conversion is the dominant contributor to extreme thermal exposure, with projected LST surpassing 45 °C in dense urban cores^{35,36}. Evidence from temperate regions such as Stuttgart further indicates that although higher building density elevates LST, building volume and surface area may exert compensatory cooling effects³⁷. Collectively, these cases demonstrate that urban form, vegetation patterns, and land-use transitions critically shape surface thermal conditions. However, they also reveal a common methodological dependence on optical thermal observations, which remain vulnerable to cloud contamination—reinforcing the need for complementary, cloud-resilient data sources such as SAR for robust characterization and prediction of urban heat dynamics^{38–40}.

Despite these advances, a critical research gap remains: SAR–optical collaborative approaches have rarely been extended beyond LST estimation to the forecasting of future thermal conditions, particularly in rapidly urbanizing Chinese cities. Existing studies predominantly focus on reconstructing current cloud-free LST—either through optical spatiotemporal fusion or by integrating SAR-derived structural indicators—but they stop short of linking SAR–optical fused datasets with predictive modeling frameworks. This limitation is especially consequential in humid subtropical regions such as Changsha, where persistent cloud cover disrupts optical time series and where rapid land-use transitions amplify the need for reliable, cloud-resilient thermal projections. To date, no study has established a framework that (i) systematically leverages SAR's all-weather scattering signals to generate continuous, cloud-free historical LST, and (ii) couples this dataset with machine-learning-based forecasting to anticipate future thermal evolution under ongoing urban expansion. Addressing this gap is crucial for enabling proactive urban climate adaptation and enhancing the accuracy of LST prediction in cloud-prone environments.

Against this backdrop, this study aims to address the identified research gap by developing an integrated, model-based SAR–optical collaborative framework for reconstructing and forecasting urban LST in cloud-prone subtropical cities. (1) Leveraging SAR's sensitivity to surface scattering properties, we establish an empirical relationship between cloud-free Sentinel-1 features (VV/VH backscatter and texture) and contemporaneous Landsat-8 thermal observations to generate a continuous, spatially complete historical LST record. This cloud-free dataset serves as the basis for training an XGBoost model that can predict future thermal conditions under various urban development scenarios. (2) By further integrating fine-scale LULC information, the proposed framework enhances the spatial–temporal continuity of thermal datasets and extends the role of SAR beyond its conventional application in classification. Overall, this study provides a cloud-resilient and transferable methodology for advancing LST reconstruction and prediction in subtropical megacities, offering actionable insights to support resilience-oriented planning, urban heat mitigation, and sustainable spatial development.

Methods

Study area and data sources

Study area

Changsha City (27°51′–28°40′ N, 111°53′–114°15′ E), situated in central southern China, serves as a strategic nexus within both the Belt and Road Initiative and the Yangtze River Economic Belt. The city is characterized by a subtropical monsoon humid climate and a distinctive horseshoe-shaped geomorphology that slopes from south to north. Mountain ranges enclosing the eastern, southern, and western boundaries—together with the only opening toward the north—create topographic conditions conducive to heat accumulation and inhibit effective thermal dispersion. The region exhibits pronounced seasonality, with concurrent rainfall and thermal patterns, and experiences long, hot, and humid summers frequently marked by extreme heat events.

Driven by rapid socioeconomic development, Changsha has undergone substantial spatial expansion over the past few decades. Urban growth has predominantly followed the north–south axis along the Xiangjiang River and the east–west development corridor. This process has resulted in extensive conversion of suburban cropland into impervious surfaces, leading to a progressive increase in high-temperature zones. Consequently, the urban heat island (UHI) effect has not only intensified but has also exhibited a trend toward decentralization,

gradually expanding toward multiple peripheral subcenters. The interplay between complex topography, regional climatic characteristics, and accelerated urbanization has thus collectively shaped the city's distinctive thermal environmental configuration. The study area comprises the urbanized core of Changsha, encompassing seven administrative divisions—Tianxin, Yuhua, Furong, Kaifu, Yuelu, Wangcheng, and Changsha County—with a total land area of 3,906.39 km² (Fig. 1). The overarching objective of this research is to elucidate the relationships between urbanization dynamics and the spatiotemporal evolution of the urban thermal environment.

Data sources

The remote sensing data used in this study include historical Sentinel-1 dual-polarization data and Landsat-8 data. The Sentinel-1 data were obtained from the Copernicus Data Space Ecosystem (Copernicus Data Space Ecosystem | Europe's eyes on Earth), and the Landsat-8 data were sourced from the United States Geological Survey (USGS.gov | Science for a changing world). The Landsat-8 data were utilized for extracting the urban LST, whereas the Sentinel-1 dual-polarization data were used for LULC classification and cloud removal from the Landsat-8 data. Information about roads, education, healthcare, and other relevant topics was sourced from OpenStreetMap (OSM). DEM data and slope and aspect information were obtained from Google Earth Engine and used for future urban LULC prediction. Multitemporal data from 2018, 2021, and 2024, along with corresponding data selected from the hottest season (August) and dates near the days with the highest temperatures and no rainfall, were incorporated into the analysis. The specific data and related details for various factors are summarized in Table 1. The selection of input variables for our LST prediction model was based on a comprehensive understanding of the physical processes governing urban surface thermal dynamics. We deliberately chose a set of features derived from multisource remote sensing data that are established indicators of surface thermal properties. To address the issue of cloud cover in our study region, we integrated SAR data (Sentinel-1), which is immune to weather conditions, into the analysis. The backscattering coefficients (VV and VH) and texture features (e.g., mean, variance, and homogeneity) extracted from SAR data using a gray-level co-occurrence matrix (GLCM) provide a robust representation of surface roughness and structure, which is crucial for assessing thermal emission. These SAR features are sensitive to urban morphology and can be effectively used to distinguish between different types of built-up areas. This multisource, multifeature approach leverages the advantages of SAR data, enabling a comprehensive and accurate representation of the urban surface. The selected variables are not only physically meaningful but also collectively capture the complex interplay among surface cover, material properties, and land surface temperature, forming a solid foundation for our predictive modeling framework.

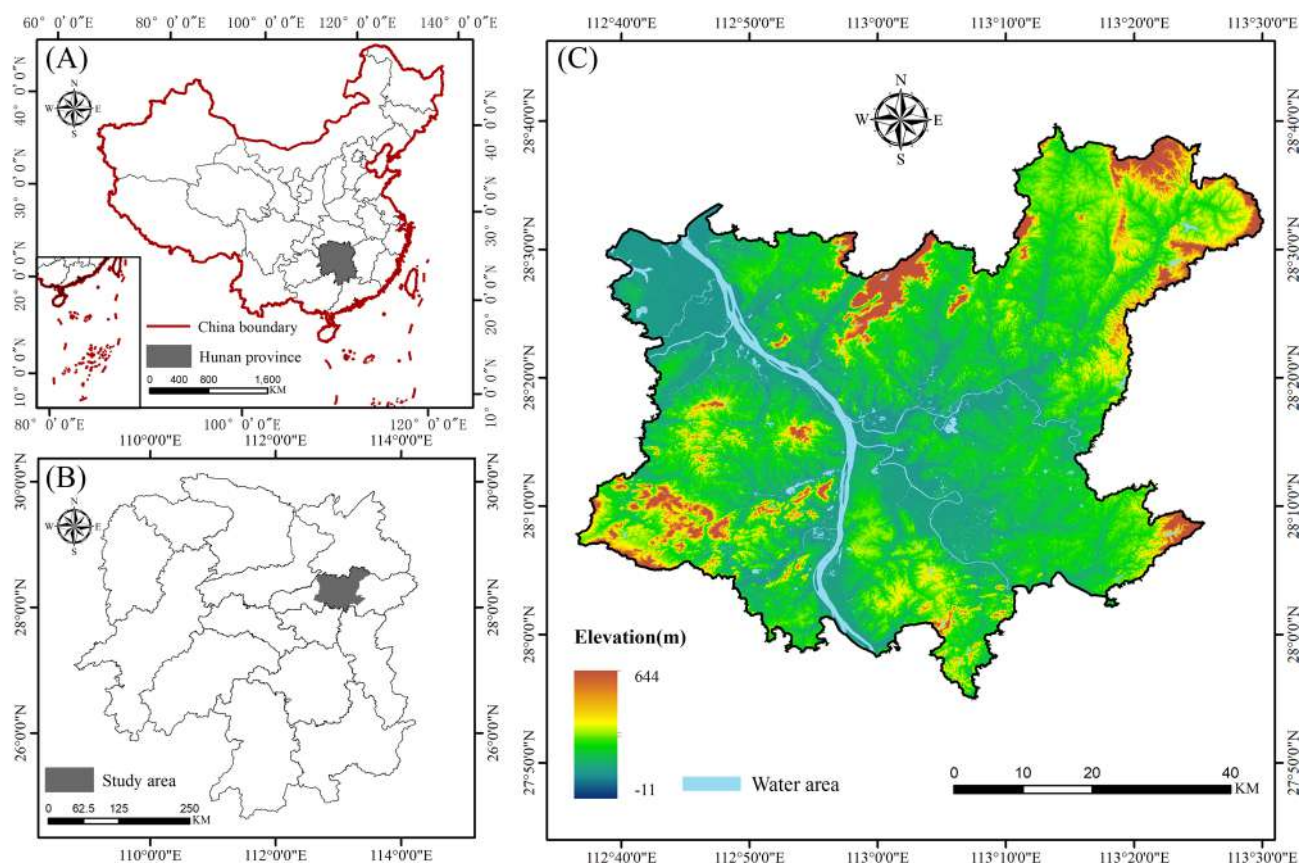


Fig. 1. (a) Location of Hunan Province in China. (b) Location of the study area within Hunan Province. (c) Study area. (created by ArcMap, version 10.8, <http://www.esri.com/>).

Data Type	Satellite Type	Acquisition	Spatial Resolution	Temporal Resolution	Source
PolSAR	Sentinel-1	Alpha	20 m	6days	Copernicus Data Space Ecosystem Europe's eyes on Earth
		Entropy			
		Anisotropy			
		VV			
		VH			
		Span			
		GLCM(n)			
Optical	Landsat-8	LST	30 m	16days	USGS.gov Science for a changing world
DEM	SRTM	DEM	12.5 m	/	Google Earth Engine
		Aspect			
		Slope			
OSM	/	Water	/	/	OpenStreetMap
		Roads			
		POI			
		Hospitals			
		Factories			
		Schools			

Table 1. Summary of data sources. Note: GLCM(n) represents the gray-level co-occurrence matrix (GLCM) at different angles. In this study, GLCMs at 0°, 45°, and 90° were used. PolSAR, Optical, and DEM data were used for LST filling prediction and LULC classification, and OSM data were utilized for LULC prediction.

The polarimetric processing of the SAR data and the extraction of associated polarimetric indicators were performed using the SNAP platform. The preprocessing workflow consisted of orbit correction, study-area subset extraction, radiometric calibration, dark noise removal, multi-looking, and generation of the C2 covariance matrix, followed by the application of the Refined Lee filter to enhance image quality further. Subsequently, the polarimetric decomposition module was employed, and the Cloude H-Alpha dual-polarization decomposition, which is appropriate for dual-polarization datasets, was applied based on the preprocessed C2 matrix. Finally, terrain correction was conducted to obtain the final polarimetric decomposition products. During processing in SNAP, auxiliary datasets such as DEM and incidence angle maps can also be generated as optional outputs to support subsequent analyses.

To analyze the three-dimensional spatial morphology evolution driven by rapid urbanization, a feature system based on all-weather SAR is constructed in this study, which is superior to the traditional optical remote sensing analysis framework. The theoretical basis for this system is as follows:

- (1) Strong correlation between 3D building morphology and LST: SAR is highly sensitive to surface scattering characteristics and can distinguish between buildings (characterized by strong double-bounce scattering) and natural surfaces (characterized by volume scattering or specular scattering) through differences in VV/VH dual-polarization scattering.
- (2) Multidimensional feature set construction: Through the selection of total power values (Span), Cloude-Pottier decomposition (CPD) parameters, and GLCM(n) features, combined with 30-meter resolution DEM data, a multidimensional index set encompassing scattering intensity, polarization characteristics, and topographic factors can be constructed.
- (3) The LULC prediction system based on the PLUS model synchronously integrates social driving factors (e.g., accessibility to healthcare, education, and industrial facilities) and ecological factors (e.g., buffer distances to water bodies and roads). All the factors are processed at a 30-meter resolution, and the coupling of multisource data enables the analysis of the mechanisms underlying the evolution of the urban thermal environment.

Technical flowchart

As shown in the technical workflow (Fig. 2), the following research methods are used in this study:

- (1) LULC classification and cloud removal: SAR feature-related indicators are established based on Sentinel-1 dual-polarization time series data from Changsha city in 2018, 2021, and 2024. The random forest algorithm is used to perform LULC classification. Moreover, Landsat-8 data are utilized to retrieve the LST, and SAR features combined with QA bands are applied for cloud removal to increase data usability.
- (2) Future urban expansion simulation: Historical LULC data and multidimensional driving factors, such as point of interest (POI) accessibility, are integrated into the framework. The PLUS model is employed to simulate urban expansion scenarios for 2027 and 2030 explicitly.
- (3) LST prediction model: SAR features, polarimetric decomposition parameters, total polarimetric power, GLCM features, natural geographic factors, and the XGBoost machine learning framework are coupled to

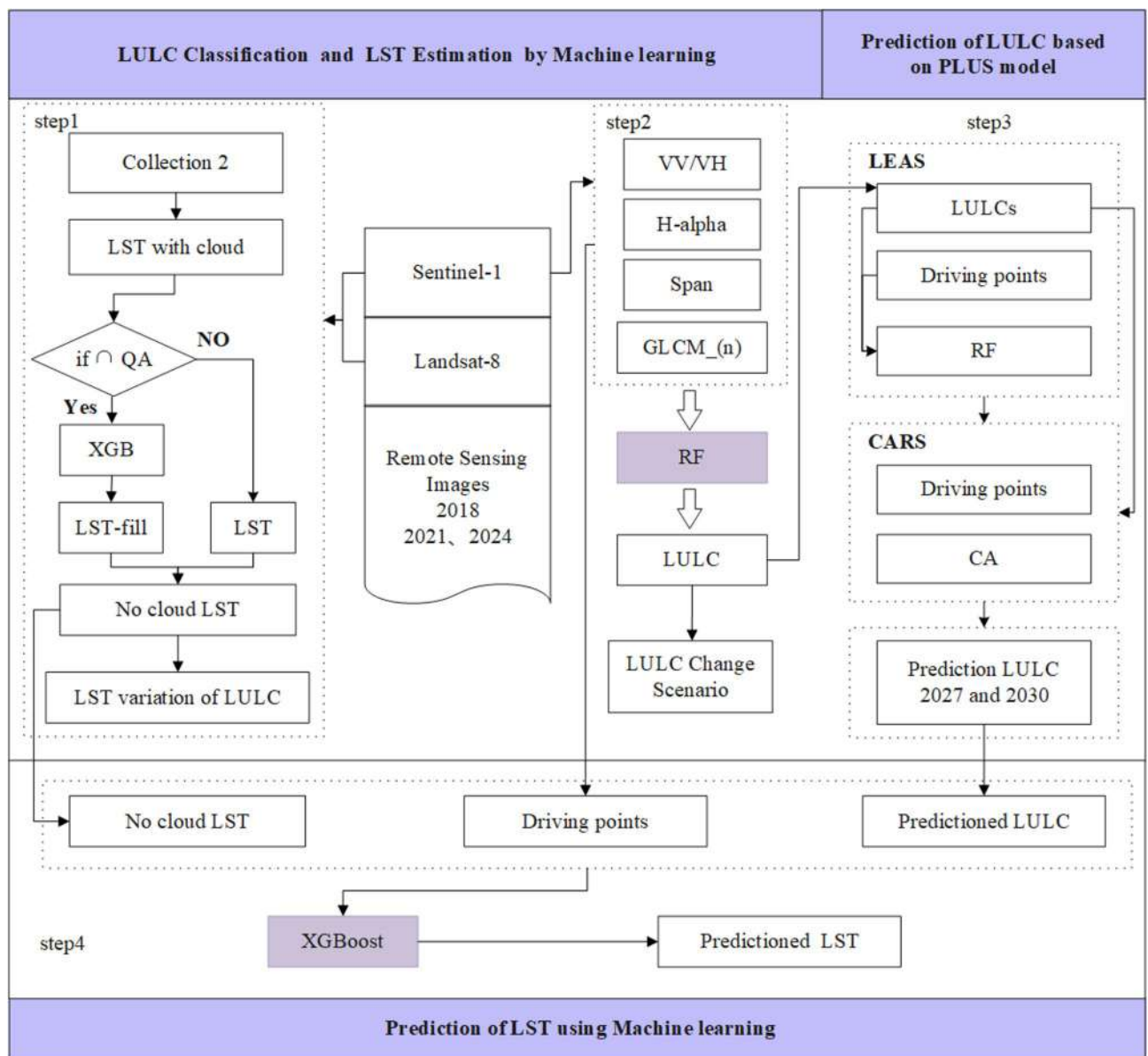


Fig. 2. Technical flowchart.

construct a temporal prediction model for the LST. The future LST conditions in the study area are predicted, and targeted urban planning recommendations and strategies are proposed based on the results.

Classification of LULC considering scattering information

PolSAR feature extraction

Sentinel-1 dual-polarization data are used in this study to extract key polarimetric feature parameters through CPD, and a GLCM analysis of the backscatter texture characteristics is performed to fully exploit the potential of radar polarimetric information in land cover classification and surface parameter retrieval. Specifically, the CPD method is applied to perform eigenvalue decomposition based on the covariance matrix to obtain polarimetric scattering entropy (H), scattering angle (α), and anisotropy (A) parameters. These parameters are used to quantify the randomness of scattering associated with targets, dominant scattering mechanisms, and energy distribution characteristics, respectively. Additionally, the total dual-polarization power (Span) is calculated to represent the overall scattering intensity of targets, which serves as the foundational input for GLCM analysis. On this basis, texture feature parameters are extracted using the GLCM, which further reveals the spatial structural characteristics of land cover. Polarimetric decomposition is used to simplify and fully comprehend complex scattering mechanisms, not only enhancing the utilization of radar polarimetric information but also providing multidimensional feature support for LULC classification and LST prediction. This approach offers a new technical pathway for remote sensing image interpretation and surface parameter retrieval.

$$H = - \sum_{i=1}^3 P_i \cdot \log_a(P), 0 \leq H \leq 1 \quad (1)$$

$$\alpha = \sum_{i=1}^3 \alpha_i P_i \quad (2)$$

$$A = \frac{\lambda_2 - \lambda_3}{\lambda_2 + \lambda_3}, 0 \leq A \leq 1 \quad (3)$$

$$Span = |VV^2| + |VH^2| \quad (4)$$

Where H represents polarimetric entropy, α represents the mean scattering angle ($^\circ$), A represents anisotropy, and λ_i represents the eigenvalues of the covariance matrix. When dual-polarization data are used, the resulting matrix is C_2 , which contains one zero eigenvalue. P_i denotes the probability derived from the eigenvalue λ_i . $Span$ represents the total polarimetric power.

GLCM extraction

The GLCM is an effective mathematical tool for analyzing surface texture heterogeneity and quantifying the frequency of gray-level co-occurrence for pixel pairs in specific directions and at particular distances. In this study, multidimensional GLCMs are constructed based on three principal directions: 0° , 45° , and 90° , to systematically extract three core parameters: contrast, angular second moment (ASM), and entropy. This forms a multidirectional texture feature combination. In urban environments, differences in building density result in significant anisotropic texture characteristics in surface cover. High-density building clusters exhibit substantial gray-level variations (high contrast) in the vertical direction (90°), whereas in the horizontal direction (0°), they display regular, repetitive patterns (low entropy). Conversely, low-density areas exhibit uniform textures in multiple directions (high ASM values). Through the integration of multidirectional texture information, the spatial patterns of buildings can be effectively captured, distinguishing between bare soil, low-rise building clusters, and high-rise building clusters. For example, the GLCM in the vertical direction highlights the vertical differences in building facades, whereas the GLCM in the horizontal direction emphasizes the regularity of roof structures. Together, these factors enhance the sensitivity of building density classification. This multidimensional analysis strategy overcomes the limitations of single-direction GLCMs in describing surface structures by integrating texture features from different directions, providing a more comprehensive feature space for urban LULC classification^{41–43}.

$$ASM = \sum_{i=0}^{L-1} \sum_{j=0}^{L-1} P_{norm}(i, j)^2 \quad (5)$$

$$Entropy = - \sum_{i=0}^{L-1} \sum_{j=0}^{L-1} P_{norm}(i, j) \cdot \log(P_{norm}(i, j)) \quad (6)$$

$$Contrast = \sum_{i=0}^{L-1} \sum_{j=0}^{L-1} (i - j)^2 \cdot P_{norm}(i, j) \quad (7)$$

Where i and j represent the gray levels (typically ranging from 0 to $L-1$, where L is the number of gray levels), d is the distance between pixel pairs, θ is the direction, and $P(i, j)$ represents the number of pixel pairs with intensities i and j separated by distance d in the θ direction.

LULC classification

In this study, a multidimensional feature set is constructed based on polarimetric decomposition parameters (H/α), GLCM(n) texture features, DEMs, latitude, and longitude. The random forest algorithm is used to classify LULC into five classes: water bodies (rivers/lakes/reservoirs), vegetation (forests), grassland (grassland/cropland/bare land), residential areas, and industrial areas. For each land cover type, 500 pixel samples are collected and divided into training and validation sets at a 7:3 ratio. Model parameters are optimized using a grid search method, and fivefold cross-validation is employed to enhance the model's generalization ability. Four different types of accuracy metrics — overall accuracy, user's accuracy, producer's accuracy, and the kappa coefficient — are assessed to evaluate the calculation accuracy of the model for different image types^{44,45}.

Land surface temperature retrieval

LST retrieval from Landsat-8 data

In this study, LST was retrieved from the thermal infrared band (B10) of Landsat-8 data to observe the spatial distribution characteristics of urban high-temperature peak areas. Landsat-8 Level-2 products, for which the B10 band has undergone radiometric calibration and atmospheric correction and surface brightness temperatures are directly provided, were used. Preprocessing was completed by the United States Geological Survey (USGS) using the radiative transfer equation, ensuring highly accurate brightness temperatures by eliminating atmospheric effects on sensor observations. In this work, the LST is obtained by scaling calculations applied to the B10 band values of the Level-2 product⁴⁶. Specifically, calibration parameters provided by the USGS are utilized, and the surface brightness temperature is adjusted in combination with surface emissivity to estimate the actual surface temperature. This approach avoids the repeated application of the radiative transfer equation while fully leveraging the preprocessed results of the Level-2 product, thus significantly improving data processing efficiency and reliability. The detailed retrieval process is illustrated in Eq. (8). To ensure data representativeness, Landsat-8 data from extremely high-temperature periods in summer (imaging day temperatures $> 35^\circ\text{C}$) are

prioritized, and cloud-free images with less than 5% cloud cover are selected to minimize the impacts of cloud occlusion and temporal differences on the identification of thermal anomaly areas.

$$ST = DN \times 0.00341802 + 149 - 273.15 \quad (8)$$

Where ST is the actual land surface temperature value (°C), DN is the pixel digital number (digital number), 0.00341802 is the multiplicative scaling factor, 149 is the additive offset, and 273.15 is the Kelvin to Celsius conversion constant.

LST interpolation method considering SAR characteristics

A significant challenge in land surface temperature (LST) modeling using optical remote sensing data—such as Landsat-8—is their susceptibility to cloud cover, which frequently disrupts observations in subtropical and humid regions such as Changsha. This leads to significant data gaps and temporal discontinuity, severely undermining the reliability of long-term urban thermal monitoring. To overcome this limitation, the synergistic fusion of SAR and optical data is implemented. Unlike optical sensors, SAR active microwave systems are capable of penetrating clouds, rain, and haze, providing consistent, all-weather, day-and-night observations. This unique capability forms the foundation of our fusion strategy, ensuring a continuous and temporally complete dataset, which is essential for uninterrupted LST time series modeling. In addition to its all-weather advantages, SAR provides a robust physical representation of urban land cover patterns that are critical drivers of LST. The sensitivity of SAR to surface geometric structure, dielectric constant, and roughness enables it to distinguish between artificial and natural surfaces based on their distinct radar backscattering signatures. For example, dense built-up areas exhibit strong and characteristic backscattering, whereas vegetation and bare soil show markedly different responses. By applying texture analysis tools such as the GLCM, SAR can further extract fine-scale textural features, enabling the effective identification and classification of varying building densities, structural types, and material compositions^{47–50}. This high-precision characterization of urban features is particularly valuable because there is a direct physical correlation between these artificial land characteristics and the spatial distribution of LST. Factors such as the building heat capacity, thermal conductivity, geometric structure, and urban canyon effect directly influence the thermal radiation and energy balance at the land surface. Consequently, the impervious surface fraction and building density, which SAR systems can accurately map, are among the most significant determinants of UHI intensity^{34,51}. By integrating cloud-penetrating SAR data, which provides continuous, high-fidelity information for key LST-driving features, with high-spectral-resolution optical data, our fusion approach not only mitigates the impact of cloud contamination but also enhances the physical interpretability of the model. This strategy fills data gaps, improves prediction robustness and accuracy, and ultimately provides a reliable, high-resolution, and continuous basis for understanding the formation mechanisms of the urban thermal environment and predicting the evolution of UHI effects.

Therefore, in this study, the LST in cloud-covered areas is reconstructed using surface scattering characteristics and other factors through regression analysis. To address issues associated with cloud-contaminated regions, Sentinel-1 backscattering features are innovatively integrated with GLCM feature parameters (angular second moment, contrast, and entropy) to establish a response model between scattering characteristics and the LST using random forest regression. First, the QA band is used to construct a cloud mask, and the study area is divided into 500 m × 500 m spatial grids in ArcGIS. The LST observation values at the grid center points and the corresponding SAR feature datasets are extracted to generate sample points. This grid size ensures that the influences of single-point features on the LST are effectively considered^{21,52}. Next, a grid search is employed to optimize the model parameters, and the training, validation, and test datasets are split at a 6:2:2 ratio⁵³. Through 5-fold cross-validation, a high-precision mapping relationship ($R > 0.84$) is obtained. Finally, the regression model is applied to fill cloud-covered areas and generate a full-area LST product. To mitigate the temporal mismatch between Sentinel-1 SAR and Landsat-8 acquisitions, we employed a nearest-date matching strategy. For each selected Landsat-8 scene, the Sentinel-1 image acquired within 16 days before or after the Landsat-8 acquisition date and closest in time was selected. This approach assumes that SAR backscatter and texture features—primarily reflecting surface structural properties—exhibit minimal variation within such a short temporal window. Detailed information on the Sentinel-1 and Landsat imagery used in this study, along with the corresponding accuracy metrics, is provided in Tables 2 and 3, respectively.

Future LULC simulation with the PLUS model

Predicting future LULC scenarios is crucial for forecasting the characteristics of urban thermal environments. Numerous models have been developed, ranging from nonspatial models such as Markov chains, which simulate changes in land use quantity but ignore spatial patterns, to spatially explicit models that simulate both quantity and spatial configuration^{54,55}. Among the latter, cellular automata (CA)-based models (e.g., CLUE-S and FLUS) and the Patch-Generating Land Use Simulation (PLUS) model have gained prominence^{56–60}. However, many of these models, including earlier CA variants, rely on simplistic transition rules or fail to fully capture the complex, nonlinear interactions among natural, socioeconomic, and spatial driving factors. The PLUS model represents a significant advancement by incorporating a more sophisticated mechanism for simulating land use expansion patterns based on multiscale driving factors and land use demand, making it a robust choice for future scenario analysis in this study^{1,35,61}.

The PLUS model framework is integrated to achieve multi-scenario urban expansion simulations, where the technical process involves the synergistic interaction of three core modules. First, a Markov chain model is employed to extract the expansion areas of different land use types between the 2018 and 2021 periods. Subsequently, samples are collected from the expanded areas, and a random forest algorithm is applied to systematically identify the driving factors underlying each type of land use expansion. This approach enables the

Year	Sensor	Acquisition Date	Scene ID
2018	Sentinel-1	2018.07.12	S1A_20180712_IW_SLC_022760_39DF
		2018.07.17	S1A_20180717_IW_SLC_022833_5424
	Landsat-8	2018.07.29	LC08_L2SP_123040_20180729_20200831_02_T1
		2018.07.29	LC08_L2SP_123041_20180729_20200831_02_T1
2021	Sentinel-1	2021.08.06	S1A_20210806_IW_SLC_039108_DBB1
		2021.08.13	S1A_20210813_IW_SLC_039210_9A8F
	Landsat-8	2021.08.22	LC08_L2SP_123040_20210822_20210827_02_T1
		2021.08.22	LC08_L2SP_123041_20210822_20210827_02_T1
2024	Sentinel-1	2024.08.14	S1A_20240814_IW_SLC_055208_F163
		2024.08.14	S1A_20240814_IW_SLC_055208_41D0
	Landsat-8	2024.08.06	LC09_L2SP_123040_20240806_20240807_02_T1
		2024.08.06	LC09_L2SP_123041_20240806_20240807_02_T1

Table 2. Details of the Sentinel and Landsat images used in the present study. Note: To fully cover the study area of Changsha city, two scenes of imagery were combined through mosaicking, compensating for the limited spatial coverage of a single scene.

Year	Train			Validation			Test		
	RMSE	MAE	R	RMSE	MAE	R	RMSE	MAE	R
2018	0.991	0.681	0.982	1.752	1.094	0.896	1.751	1.099	0.896
2021	0.997	0.651	0.972	1.767	1.031	0.841	1.779	1.038	0.841
2024	1.067	0.727	0.980	1.986	1.181	0.870	1.993	1.088	0.869

Table 3. Data filling accuracy.

quantification of development probabilities for different land-use types and the assessment of the contributions of driving factors to land-use expansion during the study period. Next, the land expansion analysis strategy (LEAS) module is used to process two periods of LULC data (2018/2021), and an RF algorithm is applied to quantify the contributions of nine driving factors (topographic factors: SRTM elevation, slope, and aspect; accessibility factors: Euclidean distance to roads/railways/water bodies; and social factors: buffer distances to hospitals/factories/schools)⁶⁰. Finally, a neighborhood weight matrix is constructed using the CA model based on multiple-class random patch seeds (CARS). By integrating random seed generation mechanisms with threshold adjustment strategies, the PLUS model enables the dynamic spatiotemporal simulation of patch autogeneration under the constraints of development probabilities. During the data preprocessing stage, the OSM source data (roads, railways, water bodies, and POIs) and SRTM terrain data are resampled to a 30 m resolution using bilinear interpolation to eliminate scale differences. Model validation is performed for the 2024 LULC distribution based on the 2018 baseline year, achieving a kappa coefficient of 0.81 and a figure of merit (FoM) index of 0.14, indicating a high degree of spatial pattern agreement between the predicted results and the actual data. The factors used in the LULC prediction are presented in Fig. 3. The parameter determination process followed established practices reported in the literature, thereby ensuring methodological consistency and reliability⁶².

Prediction of LST using the XGBoost algorithm

In recent years, machine learning has seen widespread application across diverse fields due to its superior performance in small-sample prediction and regression tasks. In the context of future LST predictions, selecting an appropriate time series forecasting method is crucial for accurate predictions. Traditional time series forecasting methods, including ARIMA and EST, perform adequately for linear relationships but often exhibit limitations when modeling nonlinear and complex patterns. In contrast, machine learning methods are better at capturing nonlinear patterns and complex features, thereby achieving more accurate predictions. XGBoost is an ensemble learning algorithm based on gradient boosting decision trees (GBDTs). The core idea of a GBDT is to progressively build a series of decision tree models, where each tree further optimizes fitting errors based on the residuals from the previous round of simulation, thus continuously reducing prediction bias. In time series forecasting, the original sequence is typically transformed to establish a supervised learning problem by using observations from past time steps as feature variables to predict future target values. This approach has yielded promising results in LST prediction^{63–65}. SAR, as an active microwave remote sensing technology, is highly sensitive to the geometric structure, dielectric constant, and surface roughness of the land surface. It can penetrate clouds and partial atmospheric disturbances, enabling all-weather observation. Its raw data (e.g., backscattering coefficients under VV and VH polarizations) and derived features—such as texture features (mean, variance, homogeneity, etc.) and polarimetric decomposition features—can effectively characterize the physical structural differences of urban underlying surfaces to distinguish and quantify various land cover types.

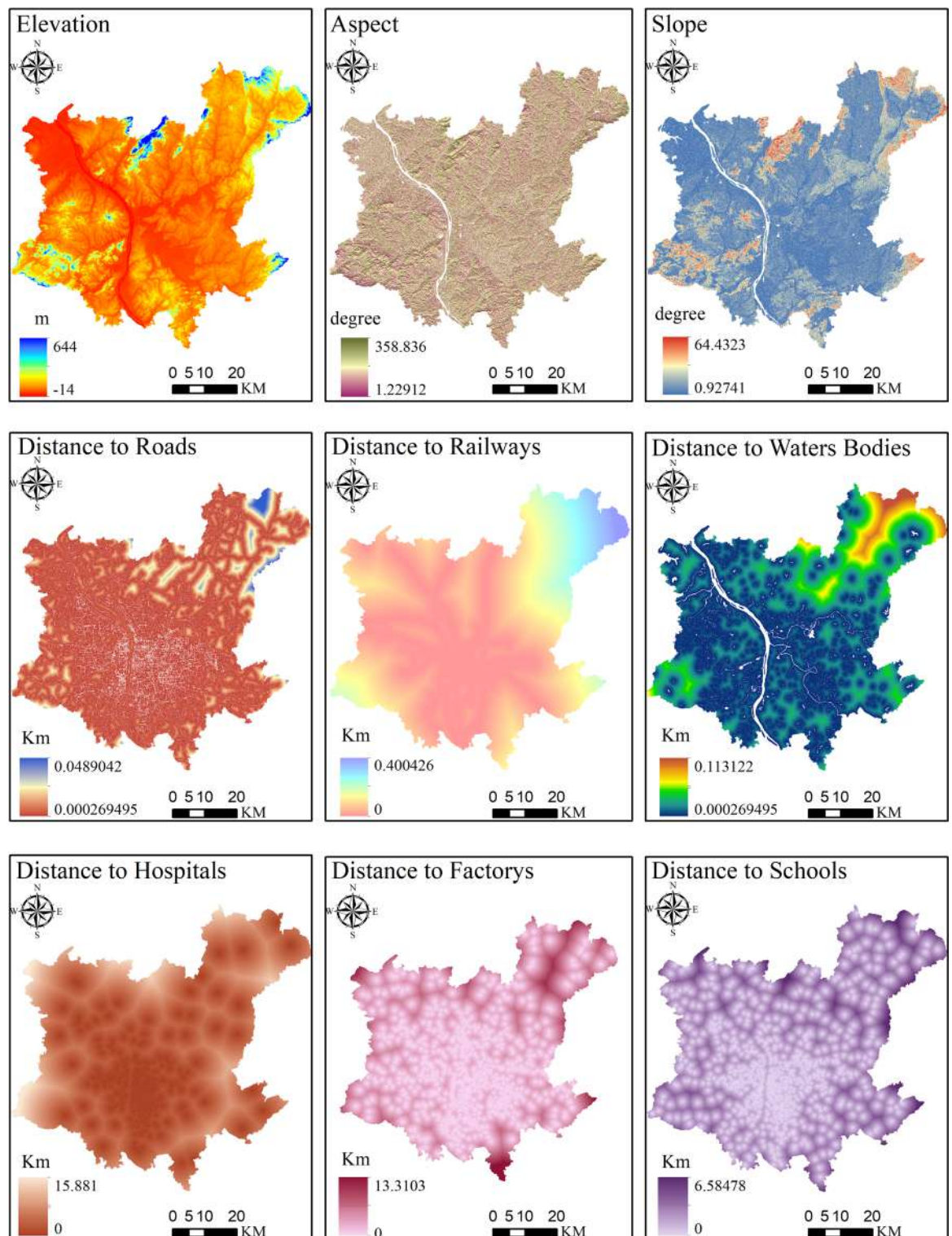


Fig. 3. Urban growth contributing factors for input into the PLUS model.

For example, high-density built-up areas typically exhibit strong backscattering and complex texture patterns, whereas vegetation and bare soil display distinct SAR responses. Given that land surface temperature (LST) is primarily regulated by physical factors such as the proportion of impervious surfaces, building density, and the thermal properties of surface materials, SAR data are ideally suited for directly sensing these key thermal driving factors. Therefore, the use of SAR and its derived features for LST prediction is grounded in a solid physical basis. Based on this mechanism, SAR data serve as the core model input and are integrated with optical remote

sensing information to construct a multisource data fusion model. This approach aims to increase the accuracy and spatiotemporal continuity of LST prediction in regions with frequent cloud cover and rainfall.

The problem of LST prediction is addressed in this study using a pixel-by-pixel prediction approach. First, a regular grid of $500\text{ m} \times 500\text{ m}$ is constructed in ArcGIS, and the LST observations at the grid center points for the years 2018 and 2021 are extracted²¹. Additionally, the analysis includes polarimetric scattering features (VV, VH, and LIA), polarimetric decomposition features (alpha, anisotropy, and entropy), total polarimetric power, GLCM features, LULC, latitude/longitude, and DEM information. These data are used for model training and validation to ensure that the predictive performance of the model meets the requirements for predicting future LST. The LST prediction model was constructed using the XGBoost regression framework. Hyperparameter optimization was performed using a grid search with five-fold cross-validation ($cv=5$), with the negative mean squared error as the evaluation metric. The search space covered three key parameters: the learning rate (0.01, 0.05, 0.1), maximum tree depth (3, 5, 7, 9), and number of boosting estimators (100, 200, 500). These parameter ranges are widely adopted in remote sensing-based thermal modeling and provide an appropriate balance between model complexity and generalization ability. In addition, an early-stopping strategy with a patience value of 50 was implemented to automatically determine the optimal number of boosting rounds, effectively reducing overfitting—particularly important given the firm spatial heterogeneity of urban thermal environments—while also improving computational efficiency.

After the model was trained using LST samples from 2018 to 2021, it was applied to predict the spatial LST distribution for 2024, and the outcomes were evaluated against the observed 2024 dataset. As shown in Fig. 4, the model exhibits strong predictive capability, achieving an RMSE of 0.9940, an MAE of 0.4714, and a correlation coefficient of 0.9819, underscoring its high reliability. The spatial comparison between predicted and observed LST indicates that the model accurately captures the overall thermal structure of the study area. Nevertheless, the enlarged sub-regions reveal slight overestimation in densely built-up and industrial zones, where surface materials and fine-scale structural variability introduce greater thermal complexity. These localized discrepancies highlight areas where future model refinement—particularly through improved representation of urban surface heterogeneity—may further enhance predictive performance.

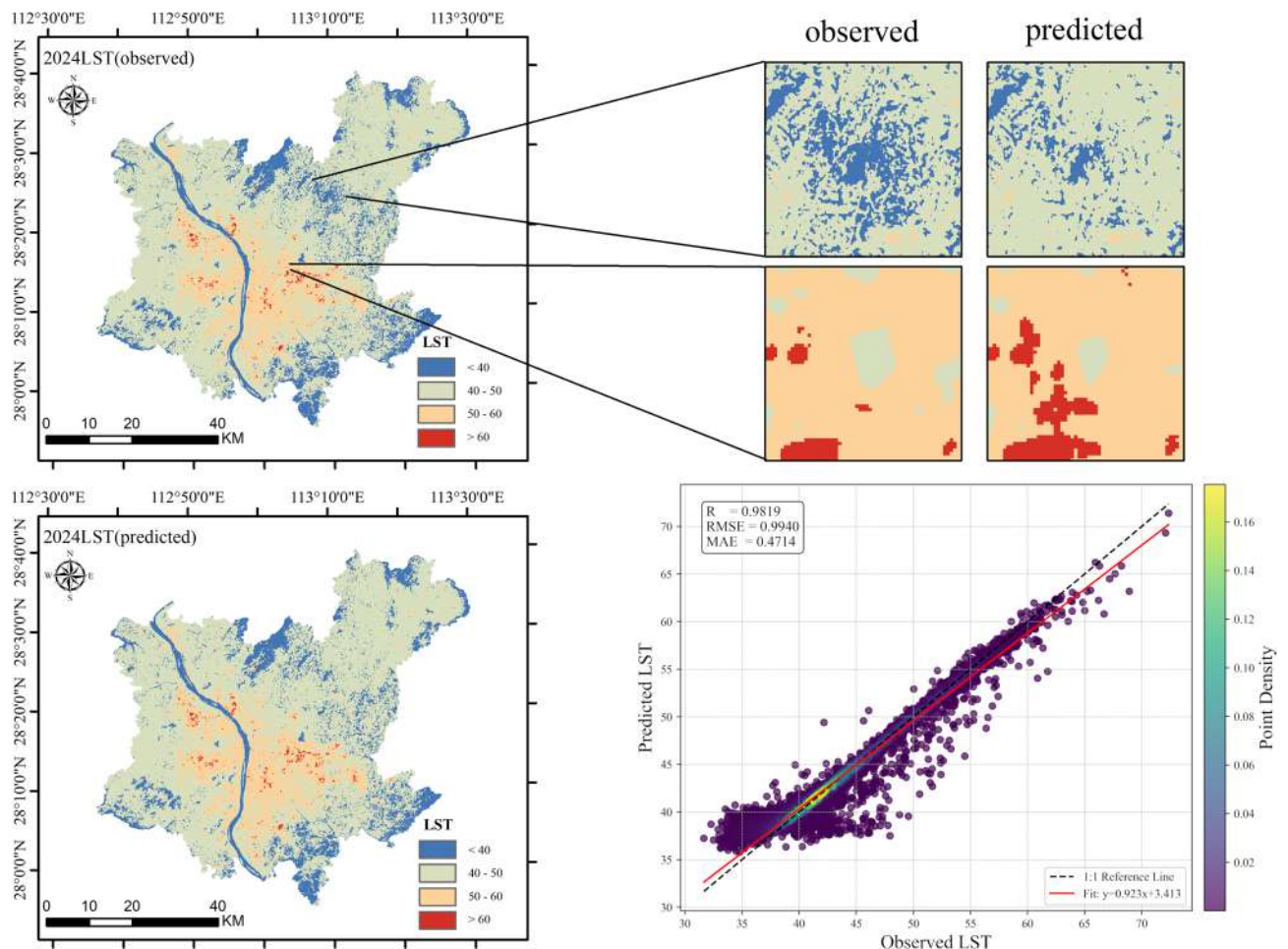


Fig. 4. Spatial and statistical comparison between observed and predicted LST in 2024. (created by ArcMap, version 10.8, <http://www.esri.com/>).

Accuracy assessment

In the LULC classification task, the kappa coefficient is used as an evaluation metric to assess classification accuracy. To evaluate the accuracy of LST regression filling and LST prediction, three metrics are considered for the final assessment: RMSE, R, and MAE. These metrics each have distinct focuses: the RMSE emphasizes significant errors, R measures the linear correlation between actual and predicted values, and the MAE provides an intuitive measure of error magnitude. By comparing the specific values of these three metrics, the overall accuracy of the results can be assessed^{66,67}.

$$\text{Overall accuracy} = \frac{\text{Total EquationNumber of corrected classified pixels (diagonal)}}{\text{Total EquationNumber of reference pixels}} * 100 \quad (9)$$

$$\text{User accuracy} = \frac{\text{Number of correctly classified pixels in each category (diagonal)}}{\text{Total EquationNumber of reference pixels in each category (row total)}} * 100 \quad (10)$$

$$\text{Producer accuracy} = \frac{\text{Number of correctly classified pixels in each category (diagonal)}}{\text{Total EquationNumber of reference pixels in each category (column total)}} * 100 \quad (11)$$

$$\text{Kappa coefficient (T)} = \frac{\text{Total EquationNumber of samples in corrected sample} - \sum (\text{col.tot} * \text{row tot})}{(\text{Total EquationNumber of samples})^2 - \sum (\text{col.tot} * \text{row tot})} * 100 \quad (12)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (LST_A^i - LST_P^i)^2} \quad (13)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |LST_A^i - LST_P^i| \quad (14)$$

$$R = \frac{\sum_{i=1}^N (LST_A^i - \bar{LST}_A) \cdot (LST_P^i - \bar{LST}_P)}{\sqrt{\sum_{i=1}^N (LST_A^i - \bar{LST}_A)^2} \cdot \sqrt{\sum_{i=1}^N (LST_P^i - \bar{LST}_P)^2}} \quad (15)$$

where LST_A^i represents the actual value of the i-th sample and LST_P^i represents the predicted value of the i-th sample. Additionally, $\bar{LST}_A$ denotes the mean of the actual values, and $\bar{LST}_P$ denotes the mean of the predicted values.

Results

Analysis of LULC changes

In this section, refined LULC classification in Changsha city over multiple years is performed by integrating polarimetric decomposition data, GLCM texture features, geographic coordinates, and DEM data. In the classification, the study area is divided into five categories: water bodies, vegetation, grassland, residential building areas, and industrial building areas (as shown in Fig. 5). According to the classification accuracy evaluation metrics presented in Table 4, the LULC classification results for 2018, 2021, and 2024 achieved high accuracy, with overall accuracy, user's accuracy, producer's accuracy, and kappa coefficient values all exceeding 0.87. These results indicate both the reliability and accuracy of the classification method. Overall, vegetation and grassland areas exhibited continuous declines between 2018 and 2024. Grassland decreased from 1,314.95 km² to 1,150.43 km² (−164.52 km², −14.30%), while forest declined from 1,241.25 km² to 1,212.89 km² (−28.36 km², −2.34%). Built-up areas expanded steadily, with residential building areas increasing from 775.33 km² to

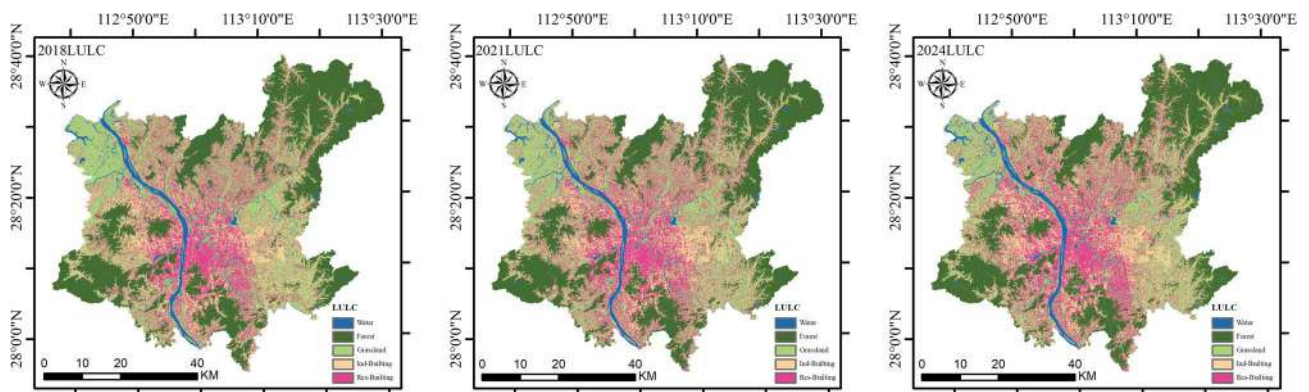


Fig. 5. LULC classification maps of the study area for 2018, 2021, and 2024. (created by ArcMap, version 10.8, <http://www.esri.com/>).

Year	User's Accuracy (%)	Producer's Accuracy (%)	Overall Accuracy (%)	Kappa Statistics (%)
2018	88.91	89.93	90.12	87.20
2021	90.53	90.42	90.67	88.31
2024	89.69	89.89	90.10	87.48

Table 4. Accuracy of LULC classification for each year analyzed based on the RF algorithm.

Class	2018	2021	2024	2018–2024	
	Area (km ²)			Area of Change (km ²)	Rate of Change (%)
Water	171.18	188.04	185.76	14.64	7.88
Forest	1241.25	1235.76	1212.89	-28.36	-2.34
Grassland	1314.95	1237.09	1150.43	-164.52	-14.30
Res-Building	775.33	842.34	870.40	95.08	10.92
Ind-Building	399.47	403.63	487.56	88.09	18.07

Table 5. Changes in LULC classes from 2018 to 2024. Note: Res-Building refers to residential building areas, and Ind-Building refers to industrial building areas.

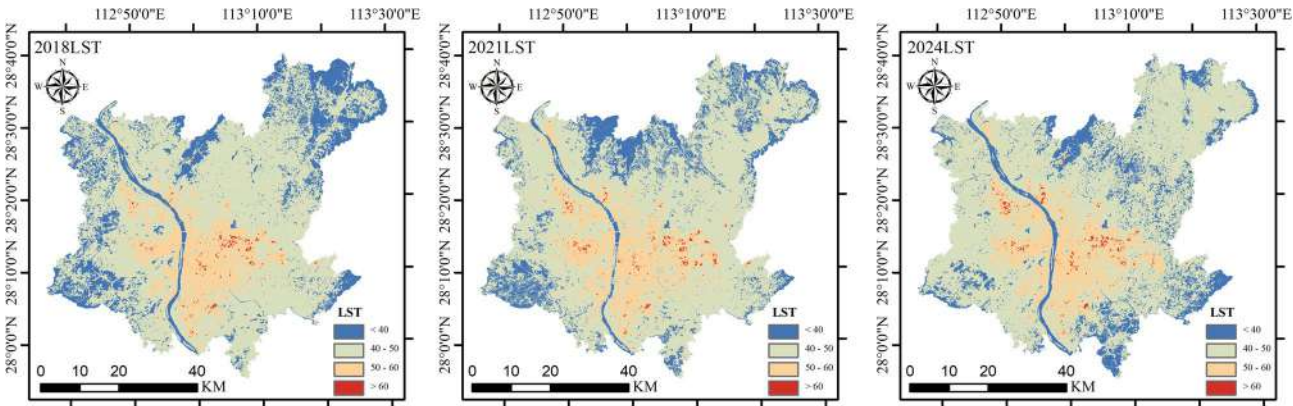


Fig. 6. Summer variation in the LST across the study area from 2018–2024. (created by ArcMap, version 10.8, <http://www.esri.com/>).

870.40 km² (+ 95.08 km², + 10.92%) and industrial building areas from 399.47 km² to 487.56 km² (+ 88.09 km², + 18.07%). Water bodies fluctuated slightly, increasing from 171.18 km² to 185.76 km² (+ 14.58 km², + 8.51%), primarily reflecting interannual variations in irrigation and river water levels. These quantified changes are summarized in Table 5.

The significant increase in industrial building areas indicates a substantial expansion of industrial zones in Changsha over the past six years. As shown in Fig. 5, industrial building areas are primarily located in the peripheral area of the city center, appearing in large clusters and expanding mainly in the east-west direction. This is because industrial activities require large tracts of land, and the lower land prices in suburban areas than in urban areas make them more suitable for concentrated industrial building zones. Additionally, industrial activities often involve pollution and high heat, making them more appropriate for the urban periphery. Moreover, the distribution of industrial building areas aligns with the layout of Changsha’s Economic Development Zone, which is consistent with the design of the city’s industrial parks. In contrast to industrial building areas, the expansion of residential building areas does not exhibit distinct geographical features. The expansion of these areas primarily involves the growth of urban central areas and rural residential zones.

Analysis of Temporal changes in land surface temperature

The use of remote sensing data can meet the requirements for large-scale LST retrieval. In this study, optical data were used to extract LST over a wide area, whereas SAR data were employed to estimate LST in cloud-covered regions through regression analysis. In the LST time series inversion process, the first step involved extracting the LST using summer Landsat-8 data from adjacent dates with peak temperatures. Second, for areas obstructed by clouds, the LST values estimated through XGBoost regression training were used to fill in the gaps. The summer LST in each year was divided into four temperature intervals: < 40°, 40–50°, 50–60°, and > 60 °C, as shown in Fig. 6. This classification facilitates the analysis of how different LULC types affect LST. The highest recorded temperatures were 76.98 °C in 2018, 81.9 °C in 2021, and 84.69 °C in 2024. The highest temperatures increased

Year	<40		40–50		50–60		>60	
	Area	Ratio	Area	Ratio	Area	Ratio	Area	Ratio
	(km ²)	(%)	(km ²)	(%)	(km ²)	(%)	(km ²)	(%)
2018	802.88	20.56	2550.59	65.31	533.25	13.65	18.77	0.48
2021	671.34	17.22	2614.69	67.08	589.10	15.11	22.77	0.58
2024	633.40	16.24	3669.76	68.49	568.70	14.59	26.68	0.69

Table 6. Distribution of different ranges of LSTs during the summer season across the study area from 2018 to 2024.

LULC	<40		40–50		50–60		>60	
	Area	Ratio	Area	Ratio	Area	Ratio	Area	Ratio
	(km ²)	(%)	(km ²)	(%)	(km ²)	(%)	(km ²)	(%)
2018								
Water	96.93	2.48	70.16	1.80	3.35	0.09	0.38	0.01
Forest	515.72	13.21	675.58	17.30	44.77	1.15	1.27	0.03
Grassland	148.07	3.79	1050.82	26.91	113.20	2.90	2.03	0.05
Res-Building	23.24	0.60	517.72	13.26	232.25	5.95	6.82	0.18
Ind-Building	17.09	0.44	234.14	6.00	139.63	3.58	8.27	0.21
2021								
Water	75.95	1.95	104.11	2.67	7.14	0.18	0.37	0.01
Forest	390.43	10.02	795.01	20.40	42.34	1.09	1.10	0.03
Grassland	129.80	3.32	985.89	25.29	115.30	2.96	2.84	0.07
Res-Building	55.66	1.43	526.33	13.50	252.65	6.48	6.40	0.16
Ind-Building	17.38	0.45	200.93	5.16	171.62	4.40	12.06	0.31
2024								
Water	104.32	2.68	74.11	1.90	6.34	0.16	0.56	0.01
Forest	334.92	8.59	835.44	21.43	35.47	0.91	1.00	0.03
Grassland	123.58	3.17	926.57	23.77	93.59	2.40	3.37	0.09
Res-Building	47.29	1.21	568.27	14.58	244.42	6.27	9.06	0.23
Ind-Building	21.06	0.54	263.22	6.75	188.84	4.84	12.70	0.33

Table 7. Cross-linkage of LST variation over different LULC classes.

over time. The peak temperatures are influenced by heat sources, with only a tiny proportion of the region experiencing extremely high temperatures at the times corresponding to temperature peaks. The use of interval-based temperature ranges effectively reflects the overall trend of temperature changes. Based on the comparative analysis of Figs. 5 and 6, and as summarized in Table 6, areas with LSTs below 40 °C decreased steadily from 802.88 km² (20.56%) in 2018 to 633.40 km² (16.24%) in 2024, representing a reduction of 169.48 km² (−4.32%). The dominant temperature range of 40–50 °C expanded from 2,550.59 km² (65.31%) to 3,669.76 km² (68.49%), an increase of 1,119.17 km² (+3.18%). The 50–60 °C range fluctuated, increasing from 533.25 km² (13.65%) in 2018 to 589.10 km² (15.11%) in 2021, before declining slightly to 568.70 km² (14.59%) in 2024. Extreme high-temperature areas (>60 °C) expanded from 18.77 km² (0.48%) to 26.68 km² (0.69%) over the same period. Spatially, high-temperature zones were primarily concentrated in central urban and industrial areas, whereas low-temperature zones were associated with forests and water bodies. Overall, the expansion of moderate- and high-temperature zones (40–60 °C) by 1,177.56 km² (+4.12%) indicates a progressive intensification of the UHI effect and a broadening of the urban thermal landscape.

Changes in LST for different LULC categories

Based on the statistical results presented in Table 7, the distribution of temperature intervals across LULC categories from 2018 to 2024 reveals pronounced spatiotemporal dynamics. For water bodies, the <40 °C area decreased from 96.93 km² in 2018 to 75.95 km² in 2021 before rebounding to 104.32 km² in 2024, while the >50 °C fraction rose slightly from 3.74 km² (0.10%) to 6.90 km² (0.17%). Forests in the <40 °C range declined markedly from 515.72 km² (13.21%) in 2018 to 334.92 km² (8.59%) in 2024, whereas the 40–50 °C interval expanded from 675.58 km² (17.30%) to 835.44 km² (21.43%), indicating a thermal shift toward higher ranges. Grasslands showed a comparable trend, with <40 °C areas shrinking and >60 °C areas gradually increasing over time. Built-up land experienced the most pronounced thermal intensification: residential buildings in the 50–60 °C interval increased from 232.25 km² to 244.42 km², while industrial buildings exhibited an even sharper rise from 139.63 km² to 188.84 km².

LULC Class	LULC Change 2018→2024 (km ²)	Change Rate (%)	LST < 40 Change (km ²)	40–50 Change (km ²)	50–60 Change (km ²)	> 60 Change (km ²)
Water	+ 14.64	+ 7.88%	+ 7.39	+ 3.95	+ 2.99	+ 0.18
Forest	− 28.36	− 2.34%	− 180.80	+ 159.86	− 9.30	− 0.27
Grassland	− 164.52	− 14.30%	− 24.49	− 124.25	− 19.61	+ 1.34
Res-Building	+ 95.08	+ 10.92%	+ 24.05	+ 50.55	+ 12.17	+ 2.24
Ind-Building	+ 88.09	+ 18.07%	+ 3.97	+ 29.08	+ 49.21	+ 4.43

Table 8. Urban land use transitions (2018–2024) and corresponding LST area Changes.

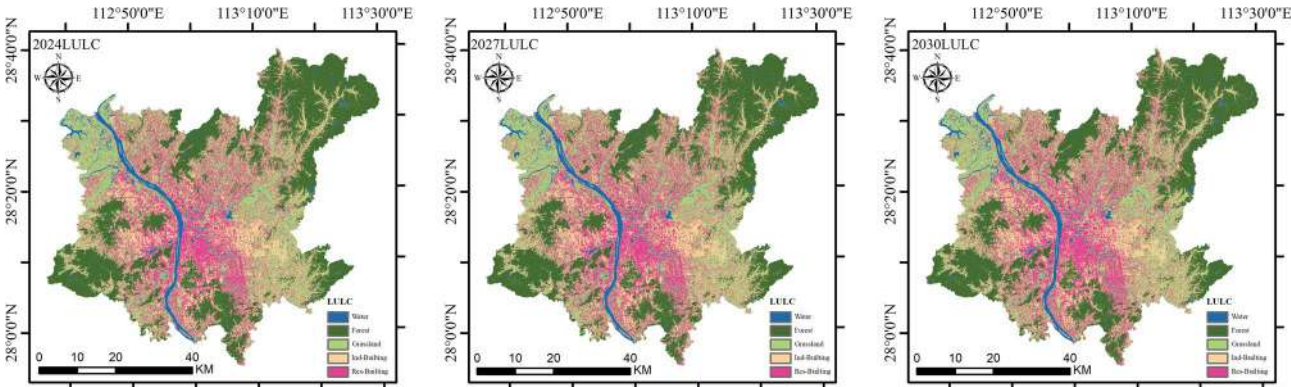


Fig. 7. Predicted LULC map of the study area in 2027 and 2030 compared with that in 2024. (created by ArcMap, version 10.8, <http://www.esri.com/>).

These patterns reflect the intrinsic thermal behavior of different land covers. Built-up surfaces—characterized by high imperviousness, low albedo, and substantial heat storage—tend to accumulate heat and promote elevated LSTs, whereas natural surfaces such as forests, grasslands, and water bodies provide cooling via evapotranspiration and higher moisture retention.

Beyond temperature distribution within each category, Table 8 further links LULC transitions from 2018 to 2024 with shifts in LST area composition, offering a more explicit quantification of how land cover changes contribute to thermal dynamics. Water bodies and built-up areas expanded during this period, while forests and grasslands declined, and these land-use transitions are reflected in corresponding LST changes. Forest loss, for example, is associated with a substantial reduction in < 40 °C areas (−180.80 km²) and a notable increase in the 40–50 °C interval (+159.86 km²), underscoring the weakening of landscape cooling capacity. Grassland shrinkage similarly leads to broad reductions across most temperature ranges. In contrast, residential and industrial expansion corresponds to consistent increases across all LST classes, particularly in the 40–50 °C and 50–60 °C intervals, highlighting the thermal impact of growing impervious surfaces. These combined results provide clear quantitative evidence that vegetation decline and urban expansion are the primary drivers of the observed warming trend across the study area.

Prediction of LULC changes

Predictions of future urban land use can inform urban planning and help address the issues associated with urbanization. Therefore, the PLUS model is used in this study to predict the LULC in 2027 and 2030. To ensure accuracy, data from 2018 to 2021, along with relevant factors, were first used to predict and validate the accuracy of the LULC data for 2024. In the PLUS model, the parameters of the random forest model were set as follows: a sampling rate of 0.01 and 200 regression trees. The evaluation metrics for the 2024 LULC prediction yielded a kappa coefficient of 0.81 and an FoM of 0.14, indicating that the model predictions were robust. The PLUS model was then used to predict LULC for 2027 and 2030, and the areas and change ratios of various land cover types were calculated (Fig. 7; Table 9). The results demonstrate that the model predictions are robust. PLUS-based projections for 2027 and 2030 suggest continued urban expansion, with residential built-up areas increasing from 870.40 km² in 2024 to 911.30 km² in 2030 (+4.70%) and industrial built-up areas expanding from 487.56 km² to 517.13 km² (+6.06%). Grasslands experienced the most significant decline, contracting from 1150.43 km² to 1095.21 km² (−4.80%), followed by forests, which declined from 1212.89 km² to 1195.64 km² (−1.42%). Water bodies remained relatively stable, rising marginally from 185.76 km² to 187.76 km² (+1.07%). Overall, these findings highlight the sustained urban growth that occurs at the expense of grassland and forest cover.

The expansion of built-up areas is characterized mainly by a gradual extension from the central urban region to the surrounding areas. During this expansion, grasslands are predominantly encroached upon, with some minor encroachment also occurring in forested areas. The increase in built-up areas in rural regions is primarily concentrated in residential zones. In contrast, the expansion of industrial zones is most prominent at the urban periphery, exhibiting significant spatial clustering characteristics. This spatial distribution pattern is influenced

Class	2018	2021	2024	2027	2030	2024–2027	2024–2030
	Area (km ²)					Rate of Change (%)	
Water	171.12	188.04	185.76	186.54	187.76	0.42	1.07
Forest	1241.25	1235.76	1212.89	1193.40	1195.64	-1.61	-1.42
Grassland	1314.95	1237.09	1150.43	1121.55	1095.21	-2.51	-4.80
Res-Building	775.33	842.34	870.40	884.73	911.30	1.64	4.70
Ind-Building	399.47	403.63	487.56	522.83	517.13	7.23	6.06

Table 9. Predicted LULC changes in the study area for 2024 and 2030.

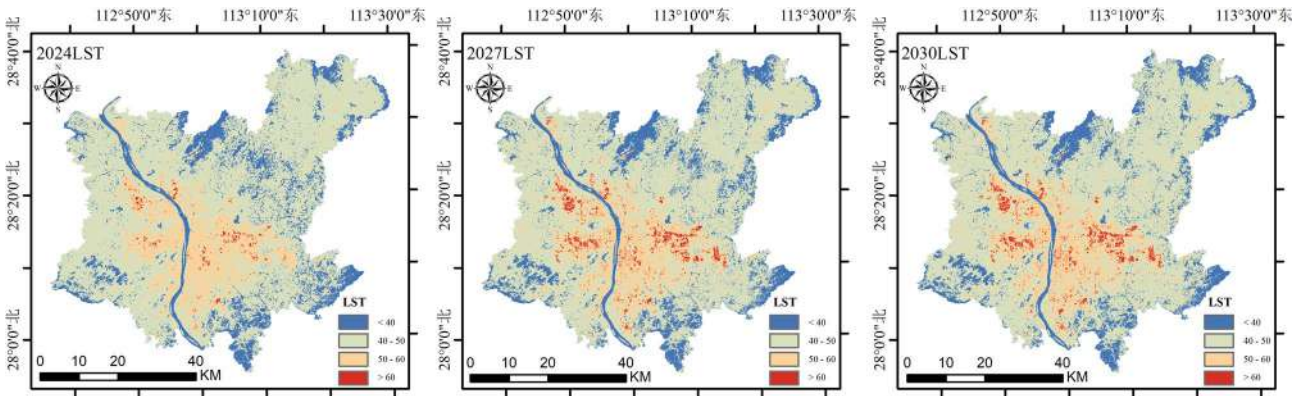


Fig. 8. Predicted LST maps of the study area in summer 2027 and 2030 compared with those in 2024. (created by ArcMap, version 10.8, <http://www.esri.com/>).

Year	< 40		40 ~ 50		50 ~ 60		> 60	
	Area (km ²)	Ratio (%)	Area (km ²)	Ratio (%)	Area (km ²)	Ratio (%)	Area (km ²)	Ratio (%)
2024	633.40	16.24	3669.76	68.49	568.70	14.59	26.68	0.68
2027	649.47	16.66	2605.96	66.85	612.68	15.72	30.08	0.77
2030	620.18	15.91	2617.62	67.15	626.78	16.08	33.61	0.86

Table 10. Predicted range of the LST over the study area for 2027 and 2030.

primarily by policy orientation. The spatial agglomeration of industries not only optimizes the synergistic effects of industrial chains but also enhances the regional economy’s overall competitiveness and its capacity for sustainable development.

Prediction of LST

As detailed in Sect. [Prediction of LULC changes](#), SAR features, GLCM textures, and natural factors were utilized to train an XGBoost model for predicting LST distributions across six districts and one county in Changsha for the years 2027 and 2030. The predictions, visualized in Figs. 7 and 8 and summarized in Table 10, reveal pronounced spatiotemporal patterns of thermal variation. The analysis indicates a clear trend of intensifying surface temperatures in conjunction with ongoing urban expansion. Core urban areas consistently exhibited the highest temperatures, with thermal anomalies extending into newly developed zones that were previously rural. Specifically, the area with temperatures exceeding 60 °C expanded from 0.6845% (26.68 km²) in 2024 to 0.7716% (30.08 km²) in 2027, and further to 0.8622% (33.61 km²) by 2030, representing a 25.9% increase over the six years. Concurrently, the 50–60 °C range expanded from 14.59% (568.70 km²) to 16.08% (626.78 km²), corresponding to a net increase of 58.07 km². In contrast, the cooler areas below 40 °C declined from 16.24% (633.40 km²) to 15.91% (620.18 km²), while the predominant 40–50 °C range remained relatively stable, fluctuating around 67% of the total area (2605.96–2617.62 km²). Spatial analysis further identified industrial building areas as the principal hotspots of extreme thermal emissions. These areas exhibited two distinct expansion patterns: (1) outward growth of existing high-temperature clusters, and (2) emergence of new thermal zones along the north–south industrial corridors of the Xiangjiang River. The concentration of extreme heat in industrial areas highlights their disproportionate contribution to the urban heat island effect, underscoring these zones as critical targets for thermal mitigation strategies.

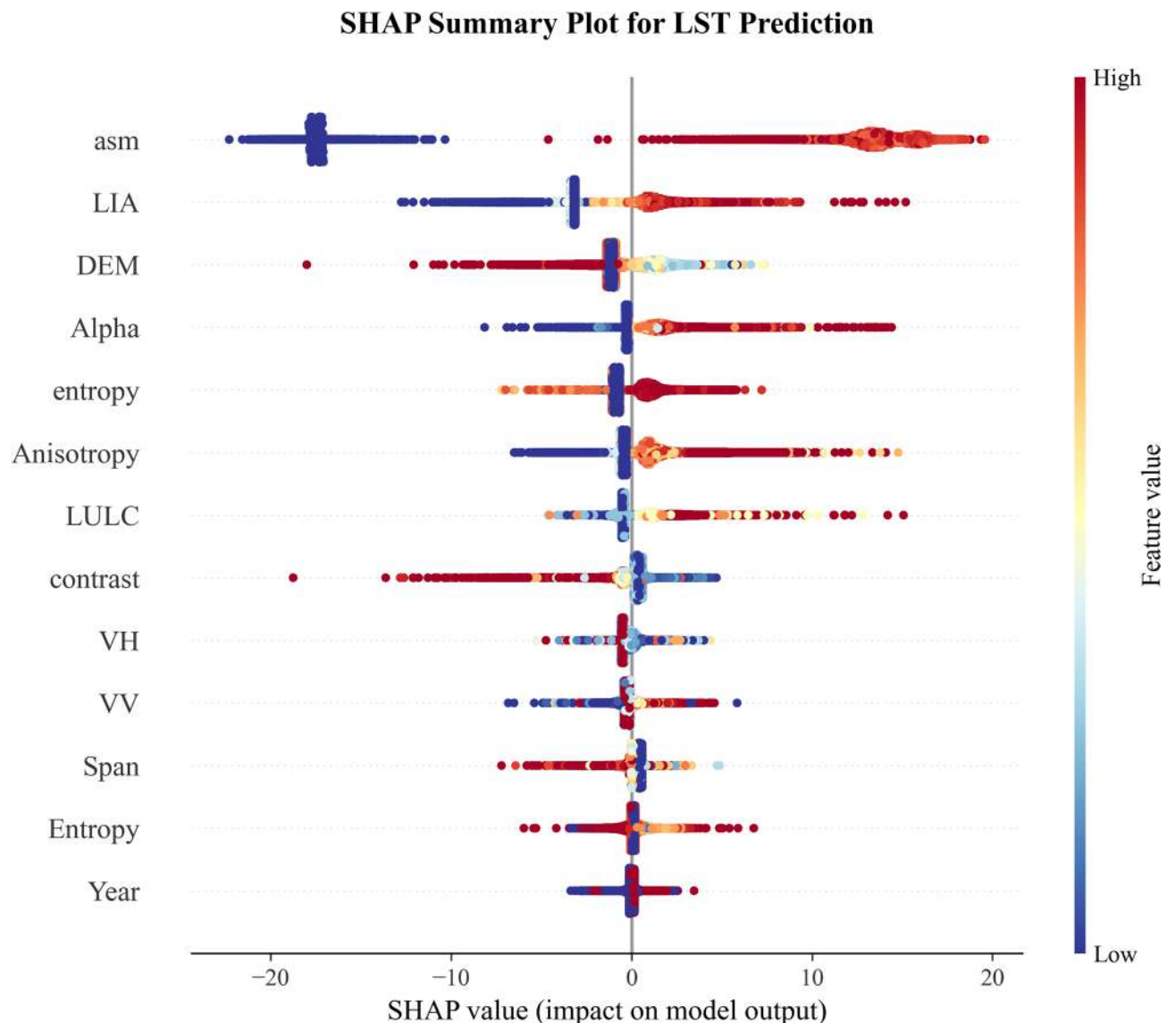


Fig. 9. SHAP summary plot of feature contributions to land surface temperature prediction.

In the SHAP summary plot for LST prediction (Fig. 9), the contributions of different features to the model output are clearly illustrated, revealing the mechanisms by which various environmental and remote sensing factors influence thermal environment modeling. In terms of feature contributions, ASM and Anisotropy exhibit SHAP values predominantly distributed in the negative range, with higher values (in red) corresponding to lower LST. This suggests that smoother surface textures or stronger directional characteristics are typically associated with lower surface temperatures, likely due to their higher thermal radiation efficiency and faster heat dissipation. In contrast, LIA, entropy (derived from the gray-level co-occurrence matrix, GLCM), VV, Alpha, and contrast show high values corresponding to positive SHAP values, indicating a strong positive driving effect. This suggests that increased surface structural complexity and heterogeneity, as well as enhanced radar backscattering intensity, contribute to higher LST. Such effects are particularly pronounced in urban built-up areas or regions with low vegetation cover and exposed land. DEM and LULC display SHAP value distributions spanning both positive and negative ranges, reflecting spatial heterogeneity in their effects on LST. For instance, high-elevation regions tend to have lower LST due to enhanced radiative cooling in thinner atmospheres. At the same time, urban built-up areas form heat islands due to high thermal capacity and poor ventilation conditions. VH also shows SHAP values distributed on both sides of zero, indicating that the vertical structure of vegetation plays a regulatory role in the surface energy balance, with different canopy structures potentially leading to either warming or cooling effects. It is worth noting that Entropy (derived from polarimetric decomposition) also exhibits high values associated with positive SHAP values, suggesting that increased randomness in scattering processes—caused by complex targets or rough surfaces—tends to promote surface warming. This finding aligns with the warming effect observed for entropy (GLCM-derived), which reflects the complexity of optical texture. By contrast, Span and Year have SHAP value distributions concentrated around zero, with near-

zero mean contributions, indicating minimal direct influence on LST prediction. This may be because other spatiotemporal explanatory variables already capture their effects. Overall, this analysis not only quantifies the relative importance of input variables in predicting LST but also uncovers the underlying physical and ecological mechanisms. It highlights the combined influence of remote sensing texture features, topographic information, polarimetric decomposition parameters, and surface cover types in thermal environment modeling, providing a strongly interpretable modeling framework for deepening the understanding of urban heat island effects and regional thermal patterns.

Discussion

Summary of key findings

This study presents an innovative application of SAR-derived features to reconstruct LST under cloud-contaminated conditions and to support LST prediction in combination with optical observations, establishing a novel methodological pathway for urban thermal studies. Unlike most previous work, which relies primarily on optical data or uses SAR mainly for LULC classification, our approach integrates SAR-derived structural, textural, and scattering metrics within a regression-based fusion framework, enabling the retrieval of thermal information even when optical data are unavailable⁶⁸. Compared with traditional optical-based LST reconstruction methods, which often deteriorate in persistently cloudy or heterogeneous urban areas, our SAR-assisted approach maintained high predictive accuracy across all land-cover types^{26,27}. The RMSE of 0.9940 in 2024, even under non-ideal observation conditions, suggests that incorporating VV and VH backscatter effectively compensates for thermal information lost in cloud-affected optical imagery. This finding aligns with previous studies that highlight the potential of SAR as a cloud-resilient thermal proxy; however, our results further demonstrate its applicability in complex urban environments with firm spatial heterogeneity.

By incorporating multiple SAR-derived variables and applying explainable machine learning, the model quantifies the relative contributions of surface structure, roughness, vegetation architecture, and land cover to urban thermal patterns. The results are consistent with global evidence: impervious expansion and industrial development strongly amplify surface heating, while vegetation and water bodies consistently provide cooling effects^{63,69}. Our findings further support the mechanistic understanding that higher surface heterogeneity, greater radar backscatter, and increased scattering randomness are associated with elevated LST. In contrast, VH-sensitive vegetation structure can modulate heat through complex biophysical interactions. The patterns observed in Changsha align closely with findings from other humid subtropical cities, such as Guangzhou and Bangkok, where urban densification drives elevated LST, while the monsoonal climate influences seasonal SUHI intensity^{32,33,37}. Comparisons with South Asian and Mediterranean cities also highlight that climate regime, vegetation phenology, and surface moisture conditions can significantly moderate SUHI magnitude and spatial distribution^{35,36}. These global parallels underscore the broader applicability of our SAR–optical collaborative framework and situate Changsha within established international patterns of urban heat dynamics. Beyond methodological contributions, the results also hold practical significance for urban planning and heat mitigation. The combined SAR–optical approach enables more reliable identification of persistent hotspots under frequent cloud cover and can therefore support targeted cooling interventions, strategic green space allocation, and climate-resilient land-use planning. The ability to retrieve LST in cloudy subtropical environments offers actionable insights for guiding heat-sensitive zoning, infrastructure design, and long-term urban climate adaptation strategies.

Research limitations

This study presents several methodological and practical limitations. First, although the PLUS model effectively simulates LULC expansion, it cannot fully capture policy-driven constraints (e.g., ecological red line protection or industrial land allocation), resulting in systematic deviations between simulated and actual development trajectories. Second, LST predictions based on XGBoost are prone to error accumulation during temporal extrapolation, which may reduce long-term forecast reliability. Third, the use of a 500 m grid helps minimize noise and computational burden but fails to represent fine-scale surface heterogeneity, potentially causing mixed-pixel effects and limiting model interpretability. Moreover, in large study areas, inconsistencies in radiometric calibration or geometric registration during multiscene mosaicking may lead to misestimation of UHI intensity. Another limitation arises from inherent temporal mismatches between SAR and optical observations. Although SAR backscattering characteristics generally vary slowly and adjacent-date matching is feasible, unavoidable discrepancies in acquisition time and sensor characteristics, along with seasonal biases and parameter sensitivity, may introduce uncertainties into the analysis. Cloud-covered area filling methods also pose several constraints: (1) filled LST values may deviate from actual surface temperatures due to missing radiative information; (2) insufficient spatial continuity may create discontinuities in temperature fields; and (3) when trained on limited samples, machine-learning-based filling risks overfitting. While the SAR–Landsat collaborative inversion framework mitigates some weaknesses of traditional multitemporal interpolation (e.g., avoiding HANTS-induced thermal smoothing), challenges persist in large-scale applications. Finally, differences in temporal and spatial resolutions between SAR and optical datasets may introduce misalignment issues that could affect the results.

Future prospects

Future research should address several key aspects to improve both the accuracy and real-world applicability of LULC and LST prediction. Technological advancements remain essential. Although current multi-source datasets enable robust modeling, limitations in spatial and temporal resolution still constrain LST reconstruction, especially under cloud-contaminated conditions. High-resolution SAR data, such as the 1 m multi-polarization GF-3 imagery, offer promising opportunities to characterize surface scattering mechanisms better and reduce

mixed-pixel effects that are common in 20 m Sentinel-1 data. Meanwhile, GF-3's short revisit cycle can substantially alleviate temporal mismatches between SAR and optical observations, thereby minimizing fusion-related uncertainties. Ground validation also warrants further attention. This study has not yet incorporated in situ temperature measurements, but ongoing field campaigns will provide essential data in future work to strengthen the completeness and reliability of LST assessments.

Policy integration represents another essential direction. The current LULC prediction framework does not fully account for planning constraints. Future research will adopt multi-scenario simulations that explicitly incorporate national spatial planning regulations—such as development boundaries, ecological redlines, and farmland protection zones—allowing more realistic projections and more substantial support for climate-resilient urban planning. Linking predicted LULC and LST patterns with heat-vulnerability assessments will further assist in optimizing green space allocation, cooling infrastructure placement, and adaptive zoning strategies.

Finally, to further enhance predictive performance, future work will incorporate advanced deep learning architectures (e.g., CNN–LSTM) and integrate socioeconomic indicators and dynamic vegetation parameters into the LST prediction framework. In addition, future studies will conduct systematic comparisons across multiple machine-learning and deep-learning algorithms to benchmark performance and improve model robustness. Future work will also include a more comprehensive uncertainty assessment, such as quantifying predictive variability and spatial uncertainty, to improve the reliability and interpretability of LST reconstruction and prediction outcomes.

Conclusions

In this study, we developed an integrated SAR–optical collaborative framework for LST retrieval and prediction, specifically designed for cloud- and fog-prone regions, thereby effectively addressing the long-standing limitations of optical remote sensing in such environments. By incorporating multipolarization SAR-derived features, the method significantly enhances data availability and improves prediction accuracy, thereby offering a novel technical pathway for urban thermal environment research. Experimental results confirm three key advantages of this approach: SAR features effectively compensate for cloud-induced data gaps, multipolarization information markedly strengthens model performance, and the framework provides a generalizable solution for regions with persistent cloud cover, such as many cities across East Asia. Beyond methodological innovation, the findings carry substantial practical implications. The predicted thermal patterns offer valuable evidence for urban planners and decision-makers to identify areas at risk of extreme heat, design more climate-resilient industrial developments, and implement effective mitigation strategies. The ability to anticipate future thermal dynamics also provides a scientific foundation for the formulation of proactive policies and urban climate management. Furthermore, our framework, which combines SAR-derived features with optical LST observations within a regression-based modeling strategy, substantially improves the accuracy and robustness of LST retrieval and prediction in cloud-affected regions, outperforming optical-only approaches. The results reveal clear spatial hotspots of extreme urban heat, concentrated primarily in rapidly expanding residential and industrial zones. These hotspot patterns provide critical evidence for prioritizing targeted cooling strategies, such as developing green infrastructure and optimizing land use. Overall, the integrated framework provides a practical and scalable decision-support tool for sustainable urban management and climate adaptation planning.

Data availability

All data used in this study are from publicly available online datasets. The SAR data were sourced from the Copernicus Data Space Ecosystem, also known as Europe's Eyes on Earth, and the Landsat-8 data were obtained from EarthExplorer. Information on roads, education, healthcare, and other factors was sourced from OpenStreetMap. Additionally, DEM data, slope, and aspect information were obtained from Google Earth Engine.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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